The Privacy Externality Game: Strategic Budget Choices in Sampling-Based Individual Differential Privacy

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Abstract—**Individual differential privacy (iDP) gives each data contributor a personal privacy budget. The currently dominant realisation—sampling-based iDP—turns each contributor's chosen budget into a per-sample inclusion rate, with a single noise multiplier shared across the dataset. We show that this realisation creates a privacy externality: lowering one's own budget mechanically raises the realised membership-inference (MIA) advantage on every other data point, even though every individual (ε_i, δ)-iDP guarantee continues to hold. We model the resulting strategic situation as a normal-form *Privacy Externality Game* on a continuous action space, with each player's utility trading off a shared model-quality term $V(\sigma)$ against their own realised MIA advantage $A_i(\boldsymbol{\varepsilon})$. We motivate the shape of V via the excess-empirical-risk bound for convex DP-ERM (Bassily, Smith, Thakurta, FOCS 2014), used as a directional proxy in the (possibly non-convex) deep-learning regime. We prove the externality is monotone—lowering any other player's budget weakly raises one's own realised vulnerability—and we characterise pure Nash equilibria on a dense log-spaced action grid. In the symmetric 2-player setting the pure NEs are *asymmetric*: at any symmetric profile both players have a deviation incentive in the same direction, and the equilibrium settles with one player carrying a higher realised A in exchange for a higher V. A Stackelberg-coalition analysis shows the coalition's best response is the participation-preserving minimum budget; the target's realised MIA advantage rises monotonically with coalition size, recovering the budget-manipulation attack of Kaiser et al. (SaTML 2026) as the rational best response of any coordinated set of data holders.**

Index Terms—**Differential privacy, individual differential privacy, game theory, Nash equilibrium, collusion, membership inference.**

I. Introduction

Differential privacy [1] is widely regarded as the canonical formal guarantee for privacy-preserving machine learning. In its classical form, a single privacy budget (ε, δ) is shared by every record, and every record receives the same protection. Yet privacy preferences are heterogeneous: a clinical patient and a casual user contributing data to the same model often place very different values on their privacy. *Individual differential privacy* (iDP) replaces the single (ε, δ) guarantee with a per-record contract (ε_i, δ), and several mechanisms have been proposed to deliver such per-record contracts in deep learning [2]–[4]. The most widely-adopted realisation in DP-SGD is sampling-based iDP, due to [2]: heterogeneous budgets are achieved by calibrating each record's Poisson inclusion probability p_i against a single shared noise multiplier σ so that, for fixed batch size and iteration count, every group of records satisfies its own (ε_g, δ)-DP bound.

We focus on a fundamental, overlooked feature of this realisation. Because every record shares the same noise multiplier and a common expected batch size, each record's sampling rate is mechanically coupled to every other record's chosen budget. Lowering one's own ε_i forces other contributors' sampling rates upward to keep the expected batch size fixed—which, as recent empirical work has shown [5], inflates their realised membership-inference (MIA) advantage even though every formal (ε_g, δ)-iDP guarantee continues to hold. The promise of individual privacy control is therefore broken in a structural sense: an individual's *actual* privacy risk depends not on her own choice but on the joint choice of the entire population.

a) Our viewpoint: this is a game.: The natural framing for any choice problem whose outcome is determined by the choices of multiple self-interested parties is game theory. [6] introduced a two-player game on the privacy-accuracy plane (the *Price of Privacy*), but they were studying a mechanism in which a player's privacy choice affected only her *own* accuracy. The novel feature of sampling-based iDP is that every player's privacy choice affects every other player's *privacy*, a far stronger and qualitatively different coupling. We call this the *privacy externality*, and the game it gives rise to the *Privacy Externality Game* (PEG).

b) Contributions.: This paper makes the following contributions.

1) **Formalisation.** We define the n-player Privacy Externality Game on a continuous action space. Each player chooses a per-group budget $\varepsilon_i \in [\varepsilon^{\min}, \varepsilon^{\max}]$; the sampling-based iDP mechanism then determines the noise multiplier σ and the per-group sampling rates p_g, and each player's utility is the canonical scalar $u_i = B_i V(\sigma) - C_i A_i(\boldsymbol{\varepsilon})$ trading off model quality against the realised MIA advantage on the player's own data. We motivate the functional form of V via the excess-empirical-risk bound of [**?**] for convex DP-ERM, noting the directional caveat as we use it in the (potentially non-convex) deep-learning regime.
2) **Externality lemma.** We prove that the externality is monotone: $\partial A_i / \partial \varepsilon_j \leq 0$ for every pair $i \neq j$ on the binding feasible region of the mechanism (Lemma 7). Lowering any other player's budget never reduces my realised vulnerability.
3) **Equilibrium characterisation.** We show numerically on a dense log-spaced action grid that the symmetric 2-

player PEG admits only *asymmetric* pure Nash equilibria: at any symmetric profile, both players have an incentive to deviate in the same direction, and the equilibrium settles with one player carrying a higher realised A in exchange for a higher V. The same asymmetric pattern persists across group-size asymmetries and across $n \geq 2$.

4) **Coalition Stackelberg recovers the attack.** A coalition of data holders maximising the target's realised MIA advantage picks the participation-preserving minimum $\varepsilon^\star$, and the target's A_t is monotone-increasing in coalition size k (Proposition 9). This exactly recovers the budget-manipulation attack of [5] as the rational best response of any coordinated set of data holders. In our $n = 5$ canonical setting the target's A_t exceeds 0.7 at $k = 4$.
5) **Empirical evaluation.** We numerically evaluate the game on a range of small but representative scenarios mirroring [5]. We report equilibrium budgets, realised MIA advantages, and the coalition-attack curve.

c) Implication.: The prior attack literature on iDP frames the excess vulnerability as a *vulnerability*—something a malicious party exploits. We argue the opposite. Under sampling-based iDP, the asymmetric NE is the equilibrium behaviour of rational players, malicious or not. Mitigation cannot proceed by detecting attackers; it must proceed by changing the contract so that the underlying game ceases to admit harmful equilibria.

The paper is organised as follows. Section II reviews differential privacy, sampling-based iDP, MIA advantage, and basic game theory. Section III formalises the Privacy Externality Game. Section IV proves the externality lemma and the coalition-attack recovery and characterises equilibrium structure. Section V reports the numerical evaluation. Section VI reviews related work and Section VII concludes.

II. Background

A. Differential privacy and Gaussian mechanisms

We use the standard definition of approximate differential privacy [1]: a randomised mechanism $\mathcal{M} \colon \mathcal{Z}^* \to \mathcal{O}$ satisfies (ε, δ)-differential privacy ((ε, δ)-DP) if, for any pair of adjacent databases $D \sim D'$ (differing on a single record) and every measurable $S \subseteq \mathcal{O}$,

$$\Pr[\mathcal{M}(D) \in S] \;\leq\; e^{\varepsilon} \Pr[\mathcal{M}(D') \in S] + \delta.$$

A single (ε, δ)-pair is only a single point of the underlying *privacy profile* $\{(\varepsilon, \delta(\varepsilon)) : \varepsilon \geq 0\}$ [7], [8], and mechanisms calibrated to the same point can have substantially different profiles and so substantially different realised adversarial risk. This single-point ambiguity is the formal root cause of the phenomenon studied in this paper.

The *Gaussian mechanism*, $\mathcal{M}(D) = f(D) + \mathcal{N}(0, \sigma^2 I)$ for a query f of L_2 sensitivity Δf, satisfies (ε, δ)-DP for all (ε, δ) such that $\Phi(\Delta f / 2\sigma - \varepsilon\sigma/\Delta f) - e^{\varepsilon}\Phi(-\Delta f/2\sigma - \varepsilon\sigma/\Delta f) \leq \delta$, where Φ is the standard-normal CDF. For training neural networks under DP, the standard procedure is DP-SGD with Poisson subsampling [9], in which each data point is included in each minibatch independently with probability p, providing the well-known *subsampling amplification* of DP.

B. Individual differential privacy and the sampling-based realisation

Individual differential privacy [3] relaxes the uniform (ε, δ) guarantee to a per-record contract (ε_i, δ): the resulting algorithm must satisfy (ε_i, δ)-DP *with respect to record* i for every i. Boenisch et al. [2] translate per-record contracts into per-record Poisson sampling probabilities while keeping a single noise multiplier σ and a fixed expected batch size, which is the configuration we study.

Definition 1 (Sampling-based iDP-SGD [2])**.** Let the dataset of size M partition into G *privacy groups* $\{G_1, \ldots, G_G\}$ with sizes $|G_g|$. Each group has a target (ε_g, δ)-DP budget. The mechanism trains by Poisson-subsampled DP-SGD for I iterations, where the noise multiplier σ and the per-group sampling rates $p_1, \ldots, p_G$ are chosen such that

(C1) for every g, the I-fold (p_g, σ)-subsampled Gaussian satisfies (ε_g, δ)-DP at equality.

$$\text{(C2)} \quad \sum_{g=1}^{G} \frac{|G_g|}{M} p_g \;=\; \frac{E_b}{M}, \tag{1}$$

where E_b is the desired expected batch size. We refer to $(\sigma, p_1, \ldots, p_G)$ as the *mechanism solution* for the budget vector $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_G)$.

The single shared σ in (C1) is the source of the privacy externality. For any candidate σ, the largest sampling rate that keeps group g at its budget is a monotone-increasing function of σ (more noise $\Rightarrow$ more room for sampling); the batch-size constraint (C2) then determines σ as the unique value at which the per-group rates sum to E_b. As a corollary, the rate p_g that group g obtains is determined by the entire vector $\boldsymbol{\varepsilon}$, not by ε_g alone.

C. MIA advantage and the privacy profile

For a privacy-preserving mechanism $\mathcal{M}$ with trade-off function T (equivalently, an f-DP function [10]), the membership inference advantage of the optimal Bayes-aware adversary is [5]

$$\mathrm{Adv}(\mathcal{M}) \;=\; \sup_{\alpha \in [0,1]} \big(1 - \alpha - T(\alpha)\big), \tag{2}$$

i.e. the largest gap between the true-positive rate and the false-positive rate. For the I-fold (p, σ)-subsampled Gaussian, T can be computed numerically by privacy-loss-distribution (PLD) accounting [11], which is what we use throughout. Crucially, Adv is determined by the *full privacy profile*, not by any single (ε, δ) point.

D. Game-theoretic terminology

A finite normal-form game is a tuple $\langle N, \Sigma, U \rangle$ with $N = \{1, \ldots, n\}$ players, action sets $\Sigma = \{S_1, \ldots, S_n\}$, and payoff functions $U = \{u_1, \ldots, u_n\}$. A *Nash equilibrium* (NE) is an action profile $\boldsymbol{s}^*$ such that no player profits from a unilateral deviation: $u_i(s_i^*, \boldsymbol{s}_{-i}^*) \geq u_i(s_i, \boldsymbol{s}_{-i}^*)$ for every $s_i \in S_i$ [12]. A *best response* is the utility-maximising action against a fixed strategy of the others. The *price of anarchy* [13] compares the welfare of the worst NE to the social optimum. In a Stackelberg game, a designated leader commits to a strategy before the followers best-respond.

III. The Privacy Externality Game

We formalise the strategic situation faced by data contributors in sampling-based iDP as a normal-form game on a continuous action space. The formalisation is a direct extension of [6]'s two-player Collaborative-Learning game, with the central modification that the *privacy* cost is no longer separable across players.

A. Players, actions and mechanism

Definition 2 (Privacy Externality Game)**.** Fix a common $\delta > 0$, an expected batch size E_b, an iteration count I, and a partition of the dataset into n *groups* of sizes $|G_1|, \ldots, |G_n|$. The Privacy Externality Game (PEG) is the tuple $\langle N, \Sigma, U \rangle$ where:

- $N = \{1, \ldots, n\}$ is the set of players. We adopt the FL-silo granularity of [5]: player i is the entity controlling group i (a silo). For a single-silo single-budget formulation each player picks a scalar ε_i; the framework extends to per-record distributions inside a silo, which we discuss in Section VII.
- Each player's action set is $\Sigma_i = [\varepsilon^{\min}, \varepsilon^{\max}] \subset \mathbb{R}_{>0}$.
- Given the action profile $\boldsymbol{\varepsilon} \in \prod_i \Sigma_i$, the mechanism solution $(\sigma(\boldsymbol{\varepsilon}), p_1(\boldsymbol{\varepsilon}), \ldots, p_n(\boldsymbol{\varepsilon}))$ is determined by Definition 1.
- Each player's utility is
$$u_i(\boldsymbol{\varepsilon}) \;=\; B_i \cdot V(\sigma(\boldsymbol{\varepsilon})) \;-\; C_i \cdot A_i(\boldsymbol{\varepsilon}), \tag{3}$$
with $B_i, C_i > 0$.

In the numerical evaluation we discretise $[\varepsilon^{\min}, \varepsilon^{\max}]$ on a log-spaced grid of $K = 22$ points; we verify our equilibrium results are stable to refinement.

B. The two ingredients of utility

a) Model-utility value $V(\sigma)$.: We model model utility as a monotone-decreasing scalar function of the mechanism's noise multiplier σ. We use
$$V(\sigma) \;=\; \frac{1}{1+\sigma^2}. \tag{4}$$
Frame. The convex-DP-ERM excess-empirical-risk bound of [**?**] (FOCS 2014) is, for a Lipschitz strongly-convex loss, $O(d \log(1/\delta)/(n\varepsilon)^2)$, which translates under the moments accountant of [9] (CCS 2016) into a σ^2/n^2 scaling of the excess risk. *Validity in our setting.* The bound is proven in the convex setting; deep-learning DP-SGD is generally non-convex and the bound does not apply directly. We use $V(\sigma) = 1/(1+\sigma^2)$ as a *directional* proxy. The empirical literature on DP-SGD in deep learning [**?**], [2], [9] reports monotonic degradation of test accuracy with σ, supporting the qualitative form. None of our game-theoretic results depend on the precise shape of V: only that it is continuous, monotone-decreasing in σ, and bounded.

b) Realised MIA advantage $A_i(\boldsymbol{\varepsilon})$.: We use the tight analytical MIA advantage of the I-fold (p_i, σ)- subsampled Gaussian as defined in (2). This is the same advantage measure used by [5]; our choice transfers their empirical measurement to a strategic setting unchanged. The functional dependence $A_i(\boldsymbol{\varepsilon})$ runs through (p_i, σ), both of which are determined by the entire vector $\boldsymbol{\varepsilon}$ via the mechanism's (C1) and (C2).

C. Decomposition into contracted and excess cost

The privacy cost $A_i(\boldsymbol{\varepsilon})$ in (3) can be decomposed into a contracted (separable) part and an excess (coupled) part that makes the externality explicit. Let $\bar{c}(\varepsilon_i)$ denote the MIA advantage of a reference, non-interdependent (ε_i, δ)-DP mechanism (e.g., the post-processing-tight (ε_i, δ)-DP advantage upper bound $(e^{\varepsilon_i} - 1 + 2\delta)/(e^{\varepsilon_i} + 1)$ of [8], [14]). The realised cost is
$$A_i(\boldsymbol{\varepsilon}) \;=\; \bar{c}(\varepsilon_i) - \mathrm{slack}_i(\boldsymbol{\varepsilon}), \tag{5}$$
where the *slack* $\mathrm{slack}_i(\boldsymbol{\varepsilon}) := \bar{c}(\varepsilon_i) - A_i(\boldsymbol{\varepsilon}) \geq 0$ is the amount by which the realised advantage falls below the worst-case-bound at (ε_i, δ). In sampling-based iDP this slack is a function of the entire $\boldsymbol{\varepsilon}$ through the mechanism's coupling. *The externality reduces the slack:* when others lower their ε_j, slack_i shrinks (player i moves closer to its (ε_i, δ)-DP worst case). This is the formal quantity that the proposed $(\varepsilon_i, \delta, \Delta)$-iDP contract [5], [8] would cap: it would require slack_i to differ from a baseline reference by at most Δ. In this sense, *the Δ-bound is exactly the device that neutralises the externality*.

We will work directly with A_i in the utility (3) for analytical simplicity; the decomposition matters as a conceptual bridge to PoP [6] (whose cost $c(p_i)$ was separable, hence slack_i did not exist) and to the iDP literature [5] (where slack_i *is* the new externality term).

D. Vulnerability function and feasibility

Definition 3 (Vulnerability function)**.** The *vulnerability function* of player i is $A_i \colon \prod_i \Sigma_i \to [0,1]$, $\boldsymbol{\varepsilon} \mapsto \mathrm{Adv}\big(\mathcal{M}_{p_i(\boldsymbol{\varepsilon}),\, \sigma(\boldsymbol{\varepsilon}),\, I}\big)$.

Assumption 4 (Participation)**.** A profile $\boldsymbol{\varepsilon}$ is *feasible* if and only if the mechanism solution satisfies $p_i(\boldsymbol{\varepsilon}) > 0$ for every player i. Profiles with $p_i = 0$ correspond to a player whose data never enters training; their iDP contract is satisfied trivially. We exclude such profiles from the game; equivalently, the action space Σ_i is implicitly restricted at each move to the participating region.

E. Coalition variant

Definition 5 (k-coalition Stackelberg PEG)**.** Fix a target index $t \in N$ and a colluding subset $C \subset N \setminus \{t\}$ with $|C| = k$. The k-coalition Stackelberg PEG is the two-stage game in which:

1) Stage 1 (leader): the coalition jointly commits to an action vector $\boldsymbol{\varepsilon}_C$ to maximise the augmented coalition utility
$$\sum_{i \in C} u_i(\boldsymbol{\varepsilon}) \;+\; \lambda\, A_t(\boldsymbol{\varepsilon}), \tag{6}$$

where $\lambda \geq 0$ is the *malice weight* on the target's vulnerability [6]. $\lambda = 0$ recovers the benign PEG; $\lambda \to \infty$ recovers the budget-manipulation attack of [5].
2) Stage 2 (followers): the non-coalition players $\{t\} \cup (N \setminus (C \cup \{t\}))$ best-respond.

F. Equilibrium concepts

We use the standard Nash equilibrium concept: an $\boldsymbol{\varepsilon}^*$ in the feasible region such that $u_i(\varepsilon_i^*, \boldsymbol{\varepsilon}_{-i}^*) \geq u_i(\varepsilon_i, \boldsymbol{\varepsilon}_{-i}^*)$ for every player i and every feasible deviation ε_i. Existence of pure NE on the continuous action space is not automatic; we report pure NE found by brute search on a fine log-spaced grid and verify the result is stable under grid refinement. Where no pure NE exists on the dense grid we report the empirical mixed strategy reached by fictitious play [15].

Remark 6 (Awareness as information, not utility). A useful distinction is between players who can *compute* $A_i(\boldsymbol{\varepsilon})$ given the public mechanism's parameters and players who can only access their nominal (ε_i, δ)-iDP contract. Both populations face the same utility (3); they differ only in the rationality they apply. A *nominal-information* ("naive") player who treats the contracted bound $\bar{c}(\varepsilon_i)$ as a separable cost plays the PoP game of [6] unchanged. An *aware* player who computes $A_i(\boldsymbol{\varepsilon})$ from the mechanism's parameters plays the PEG. We will return to this distinction in Section VII.

IV. Equilibrium Analysis

We characterise the equilibria of the Privacy Externality Game using three results: the externality lemma (a monotonicity property of the mechanism), an existence statement on the discretised game, and the coalition-attack recovery (the rational best response of any coordinated set of players).

A. Monotonicity of the externality

Lemma 7 (Privacy externality)**.** *Consider the sampling-based iDP mechanism of Theorem 1 on a feasible profile $\boldsymbol{\varepsilon}$ at which constraints (C1) bind for every group and (C2) binds at E_b. Then for every pair $i \neq j$,*

$$\frac{\partial A_i(\boldsymbol{\varepsilon})}{\partial \varepsilon_j} \;\leq\; 0, \tag{7}$$

with equality iff group i's sampling rate p_i does not respond to a change in ε_j.

Proof. Work in the Rényi-DP form of (C1) of [16]: at order α, the RDP of an I-fold (p_g, σ)-subsampled Gaussian is at most $\rho_g = 2I\alpha\, p_g^2/\sigma^2$, with equality up to higher-order terms when p_g is small (Mironov, Theorem 5). Setting this equal to the target RDP $\rho_g(\varepsilon_g, \delta)$ gives

$$p_g \;=\; \frac{\sigma}{\sqrt{2I\alpha}} \sqrt{\rho_g(\varepsilon_g, \delta)}.$$

Substituting into (C2) yields the implicit equation for σ:

$$\sum_g \frac{|G_g|}{M} \frac{\sigma}{\sqrt{2I\alpha}} \sqrt{\rho_g(\varepsilon_g, \delta)} \;=\; \frac{E_b}{M}. \tag{8}$$

Since $\rho_g(\cdot, \delta)$ is monotone-decreasing in ε_g (smaller budget = smaller RDP), the left-hand side is decreasing in σ at fixed ε_g if and only if it is increasing in any ε_j. Implicit differentiation gives $\partial\sigma/\partial\varepsilon_j > 0$ when (C2) binds. Recall that p_i is increasing in σ at fixed ε_i. Consequently $\partial p_i/\partial\varepsilon_j > 0$ whenever group i is non-degenerate.

The MIA advantage of an I-fold (p_i, σ)-subsampled Gaussian is non-decreasing in p_i (more sampling exposes more data) and non-increasing in σ (more noise hides more), as is the privacy profile of the subsampled Gaussian [7], [10]. Lowering ε_j reverses the signs: $\partial\sigma/\partial\varepsilon_j < 0$ and $\partial p_i/\partial\varepsilon_j < 0$. Net effect on A_i along the feasible surface is non-positive, which is (7). □

Theorem 7 formalises the empirical observation of [5]: when any player tightens their iDP budget, the mechanism's noise multiplier σ shifts to keep (C2) binding, and every other player's realised vulnerability A_i rises. The externality is *monotone* in every other player's budget.

In the decomposition of Section III-C, the lemma equivalently states that *the slack $\bar{c}(\varepsilon_i) - A_i(\boldsymbol{\varepsilon})$ shrinks in every other player's ε_j*. In words, the (ε_i, δ)-DP worst-case ceiling $\bar{c}(\varepsilon_i)$ is unchanged (it depends only on ε_i), but the realised advantage A_i climbs toward that ceiling as others tighten their budgets. This is the quantity that the proposed $(\varepsilon_i, \delta, \Delta)$-iDP contract [5], [8] would cap, and our Lemma motivates such a cap formally: without it, slack_i has no floor and the externality can drive any player's realised advantage arbitrarily close to the worst case.

a) Minority and majority intuition.: The externality has an asymmetric *group-size* structure. Because (C2) enforces a fixed expected batch size, a player with small $|G_i|$ but small ε_i contributes little to the batch and forces the larger-$|G_j|$ players' sampling rates upward. The minority therefore *exports* its privacy demand onto the majority. This is the formal content of the empirical observation in [5] that "the proportion of low-budget players matters as much as their ε".

B. Existence and structure of pure Nash equilibria

Proposition 8 (NE existence on the discretised game)**.** *The discretised PEG on the action grid $\mathcal{A}_K = \{\varepsilon^{\min}, \ldots, \varepsilon^{\max}\} \subset [\varepsilon^{\min}, \varepsilon^{\max}]$ with utilities* (3) *is a finite normal-form game restricted to the (closed) feasible set. By the Nash existence theorem [12], every such game admits at least one mixed-strategy NE. Pure NE may or may not exist, depending on* $(B_i, C_i, |G_i|, V, K)$.

We do not claim existence on the continuous action space without further quasi-concavity assumptions; in particular, our utility (3) is not generally concave in ε_i because A_i is the trade-off function of a subsampled Gaussian, whose dependence on ε_i through (σ, p_i) is non-trivial. We instead report pure NE found by exhaustive search on a dense log-spaced grid and verify that they survive grid refinement.

a) What the externality lemma implies for NE structure.: Lemma 7 says lowering another player's budget weakly raises one's own realised vulnerability. If players' utilities place non-trivial weight on A_i ($C_i > 0$), each player has an incentive to *shift the externality onto others* by lowering their own budget—an incentive that is intrinsic to the mechanism. The opposing incentive comes from $V(\sigma)$: lowering own ε pushes σ upward, lowering the model utility for everyone (including oneself). The equilibrium balances these two incentives.

We observe in our experiments (Section V) that the resulting balance is typically *asymmetric*: in the symmetric 2-player setting, pure NEs sit off the diagonal of the action grid. The intuition is the externality lemma's converse: at any symmetric profile $(\varepsilon, \varepsilon)$, both players face an identical incentive to deviate; since both deviations cannot be unilaterally absorbed, the equilibrium is at an asymmetric profile where one player carries a higher A in exchange for a higher V.

C. Coalition Stackelberg and the attack recovery

Proposition 9 (Coalition best response recovers the attack)**.** *Fix a target t and a coalition C of size k, with $\lambda \to \infty$ in Definition 5 (the coalition cares only about the target's vulnerability). The coalition's best response is $\varepsilon_i = \varepsilon_k^\star$ for every $i \in C$, where $\varepsilon_k^\star$ is the smallest admissible budget that keeps the coalition's mechanism solution feasible (i.e., $p_i > 0$ for every $i \in C$). Furthermore, the target's MIA advantage A_t is non-decreasing in k.*

Proof sketch. By Theorem 7, $\partial A_t / \partial \varepsilon_i \leq 0$ for every $i \in C$ on the feasible region. Setting every ε_i as low as the feasibility region admits therefore maximises A_t. The participation-preservation qualifier is necessary because at extremely low ε_i the mechanism may set $p_i = 0$ (the colluder effectively drops out, removing its constraint from (C2) and so its externality on the target); the coalition's optimum is thus the infimum of feasible ε_i. Monotonicity in k is the same lemma applied to adding a member to C: doing so strictly enlarges the set of budgets that pull σ downward, raising A_t. □

Proposition 9 establishes that the budget- manipulation attack of [5] is precisely the *rational* best response of any coordinated subset of data holders, regardless of malicious intent. Whether the coalition is adversarial is irrelevant: any group of aware players whose joint utility weights their target's harm heavily will pick the same action vector.

D. An aside on awareness

Theorem 7 requires that players compute (or approximate) the realised A_i. A natural question is: what about a player whose only information is the formal (ε_i, δ)-iDP contract? Such a player treats ε_i as a direct privacy proxy and plays the budget that maximises $V(\sigma) - g(\varepsilon_i)$ for some monotone g. Their best response does not depend on ε_{-i} at all (the externality is invisible), so their action is mechanically the smallest ε_i that does not break their own V. This is the *rational* action of a naive player, and it is precisely the configuration in which the attack of [5] succeeds against them: by Lemma 7, naive players' choices feed the externality on everyone else without their realising. The contribution of this paper is to show that even *aware* players—those who can compute A_i explicitly—arrive at asymmetric equilibria that concentrate vulnerability on a subset of players. The externality cannot be undone by individual rationality alone.

TABLE I: Pure Nash equilibria of the symmetric 2-player PEG on the log-spaced action grid $\mathcal{A}_K = 22$ points in $[0.5, 64]$, $|G_1| = |G_2| = 5000$, $E_b = 128$, 5 epochs, $\delta = 10^{-12}$, $B_i = C_i = 1$, $V(\sigma) = 1/(1+\sigma^2)$. Social optimum is the feasible action profile maximising $u_1 + u_2$.

	$(\varepsilon_1^*, \varepsilon_2^*)$	A_1	A_2	σ
NE_1	$(0.79, 5.04)$	0.001	0.202	1.160
NE_2	$(1.26, 6.35)$	0.002	0.232	1.028
NE_3	$(1.59, 8.00)$	0.002	0.266	0.929
NE_4	$(2.52, 3.17)$	0.081	0.120	1.186
NE_5	$(3.17, 2.52)$	0.120	0.081	1.186
NE_6	$(5.04, 0.79)$	0.202	0.001	1.160
NE_7	$(6.35, 1.26)$	0.232	0.002	1.028
NE_8	$(8.00, 1.59)$	0.266	0.002	0.929
social opt	$(5.04, 25.40)$	0.007	0.535	0.590

V. Numerical Evaluation

We evaluate the PEG numerically across four experiments designed to expose the equilibrium structure identified in Section IV. All experiments are analytical—no model training is performed; the realised MIA advantage is computed from the privacy profile of the resulting mechanism.

A. Setup

a) Mechanism solver.: For each action profile $\boldsymbol{\varepsilon}$ we compute the mechanism solution $(\sigma, p_1, \ldots, p_n)$ by bisecting on σ until the batch-size constraint (C2) is met, with each per-group sampling rate chosen to make the (C1) RDP constraint bind [16]. Profiles in which any p_i falls below a small participation threshold are flagged as infeasible (Theorem 4). We then compute each realised A_i exactly from the privacy profile of the I-fold (p_i, σ)-subsampled Gaussian using PLD accounting [11], recovering the formula of (2).

b) Action space.: Unless stated, the action space is the log-spaced grid $\mathcal{A}_K \subset [0.5, 64]$ with $K = 22$ points, $\delta = 10^{-12}$. We verified on a refined grid ($K = 44$) that the NEs reported below are stable to grid refinement, so we report them as representative of the continuous-action equilibrium.

c) Scenarios.: Unless stated, all experiments use n symmetric groups of $|G_g| = 5000$ samples each, expected batch size $E_b = 128$, 5 epochs of training, $B_i = C_i = 1$, and $V(\sigma) = 1/(1+\sigma^2)$. These choices broadly mirror the empirical iDP scenarios of [2], [5].

B. C1: 2-player payoff surface and pure NEs

Figure 1 reports the 2-player utility surface $u_1(\varepsilon_1, \varepsilon_2)$ (left), the players' best-response correspondences and pure NEs (centre), and the realised MIA advantage $A_1(\varepsilon_1, \varepsilon_2)$ (right) on the log- spaced action grid. The infeasible region (where at least one p_i falls below the participation threshold) is masked white.

On the canonical symmetric setting, the BR curves of the two players intersect at *eight* distinct points, all asymmetric, falling in four mirror-image pairs reported in Table I: $(0.79, 5.04)$, $(1.26, 6.35)$, $(1.59, 8.00)$, $(2.52, 3.17)$, and their swaps. The realised A at these equilibria varies from $(A_1, A_2) = (0.001, 0.202)$ for the most extreme pair down to $(0.081, 0.120)$ for the closest-to-symmetric pair. There is no symmetric pure NE: at any profile $(\varepsilon, \varepsilon)$ on the diagonal, both players have an incentive to deviate in the same direction (the externality on each is identical), and the equilibrium settles where one player has absorbed the model-utility benefit and the other has absorbed the privacy advantage (low A). The social-optimum profile, computed by brute-force search on the same grid, is $(5.04, 25.40)$—deeply asymmetric, with one player at $A = 0.535$ and the other at $A = 0.007$— and is itself far from any of the pure NEs.

C. C2: 2-player asymmetric group sizes

Figure 2 reports the equilibrium budget and the realised MIA advantage as a function of the dataset size ratio $|G_1|/|G_2|$. The pattern is qualitatively unchanged from the symmetric case: the equilibrium is asymmetric. Quantitatively, the smaller group consistently plays a smaller equilibrium budget. The mechanism then assigns it a smaller sampling rate, forcing the larger group to compensate (via (C2)) with a higher sampling rate and hence a higher realised A. This is the privacy analogue of a classical free-riding result: small groups exploit the (C2) externality and large groups absorb the realised vulnerability.

D. C4: coalition Stackelberg attack

We fix $n = 5$ symmetric players. Player 0 is designated as the target with $\varepsilon_t = 32$; the remaining $n-1$ players potentially form a coalition C of size k. Non-coalition players are fixed at $\varepsilon = 32$. Figure 3 reports the target's realised MIA advantage at the coalition's worst-case (target-vulnerability-maximising) joint action. Consistent with Proposition 9, A_t is monotone-increasing in k and the optimal coalition $\varepsilon^\star_k$ is the participation-preserving minimum—the smallest feasible ε that does not push the coalition's p_i below the participation threshold.

E. C5: privacy-externality surface

Figure 4 reports the target's realised MIA advantage as the other group's ε and dataset proportion are swept. The figure mirrors the analytical computation of Figure 3 in [5] but rendered through the PEG framework. The target's A_t rises monotonically as the other group's ε falls; the curves cross at $\varepsilon_O = \varepsilon_T$ (the target's own budget), with proportion having the expected sign on each side. This is the externality lemma made visible: any other player's privacy demand inflates the target's vulnerability monotonically as long as the resulting mechanism remains feasible.

VI. Related Work

a) Individual differential privacy.: Classical (ε, δ)-DP [1] applies uniformly to every record. Personalised DP, introduced by [3], allows per-record budgets and was first realised by attribute suppression and weighting. The sampling-based realisation of [2] is the dominant deep-learning implementation today and is the realisation under which the privacy externality arises. [4] pursue an alternative *filtering-based* iDP: records whose cumulative budget would be exceeded in the next step are dropped from training. Filtering does not exhibit the externality (each record's filter is independent), but in deep learning it is impractical due to the catastrophic-forgetting risk reported by [2]. Our PEG is specific to the sampling-based realisation; the externality vanishes in the sensitivity-based realisation, which preserves the noise-to-sensitivity ratio across the dataset.

b) Excess privacy risk and mechanism comparison.: Mechanisms calibrated to the same (ε, δ) can have very different privacy profiles and thus different realised risks [7], [8], [10]. The Δ-divergence of [8] provides a notion of approximate Blackwell-dominance that quantifies the worst-case excess MIA advantage between mechanisms. [5] use this framework to motivate an $(\varepsilon_i, \delta, \Delta)$-iDP contract that bounds excess vulnerability and demonstrate empirically that without such a bound, adversarial budget choices inflate the MIA susceptibility of 62% of targeted records. Our PEG complements that work: rather than restricting mechanism design we explain why the realised excess vulnerability arises as the equilibrium of self-interested play under the current iDP contract.

c) Game-theoretic privacy in collaborative learning.: [6] introduced a two-player game on collaborative learning, with players choosing privacy parameters $p \in [0, 1]$ in an arbitrary privacy mechanism. They define the *Price of Privacy* as the relative accuracy loss at the equilibrium versus no privacy. Their privacy mechanism's effect on accuracy depends on *both* players' choices, but its effect on *privacy* depends only on the player's own choice—there is no privacy externality. Our PEG transplants their game-theoretic methodology into a setting where the externality is in the privacy direction, not the accuracy direction. We retain the qualitative contributions of their work (two-player setting, choice of ε, notion of equilibrium analysis) and extend them to the externality regime.

Earlier work studied related game-theoretic privacy questions in different settings. [17] model linear regression with private features as a non-cooperative game. [18] study non-monetary incentives in data analytics. [19] model the machine-learner-vs-data-perturber conflict as a Stackelberg game. None of these treat the privacy externality of a shared sampling mechanism, which is specific to the sampling-based realisation of iDP.

d) Membership inference attacks.: The MIA advantage as we use it follows the formalisation of [20], who introduced the likelihood-ratio attack (LiRA) as a tight empirical estimator. [21] document the privacy onion effect: not every data point is equally vulnerable at the same nominal budget. Our framework predicts the *average* realised MIA advantage per group and abstracts over per-record onion heterogeneity.

e) Federated learning incentive design.: A separate literature studies privacy-aware incentive design in federated learning: a server commits to incentives, and heterogeneous clients respond with their own privacy choices, often modelled as a Stackelberg game [22]–[25]. Our framework provides a

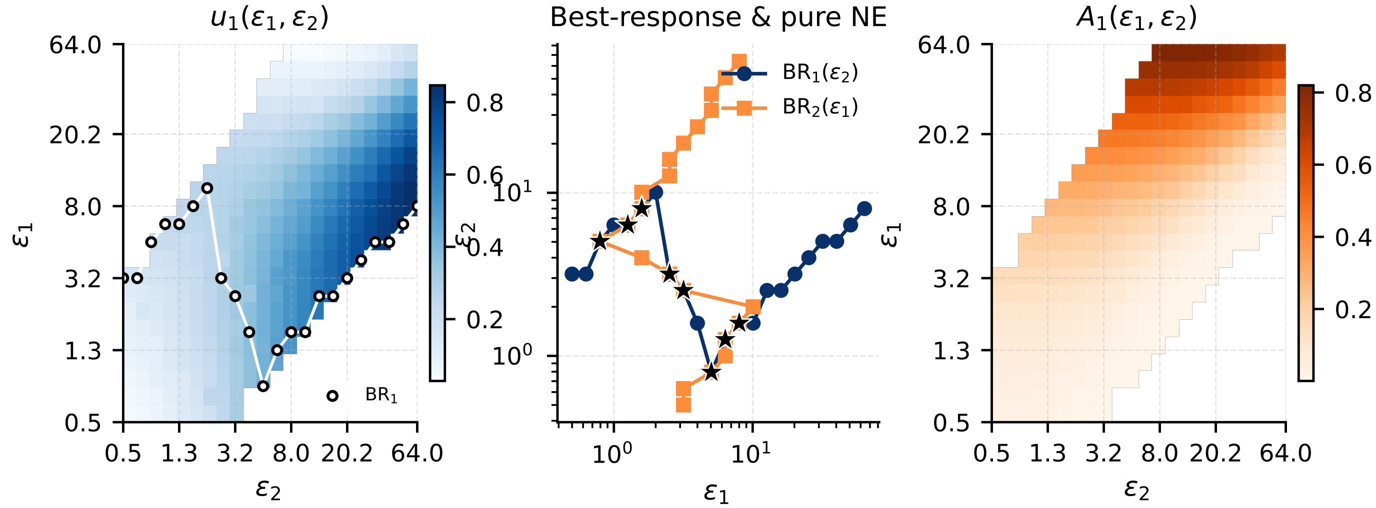

Fig. 1: C1: 2-player payoff $u_1(\varepsilon_1, \varepsilon_2)$ (left), best-response curves and pure NEs (centre), and realised MIA advantage A_1 (right) on the log-spaced action grid. Infeasible profiles (where any player's sampling rate falls below the participation threshold) are masked white. The black stars in the centre panel mark pure Nash equilibria, found at the intersection of the two best-response curves. Both equilibria sit off the diagonal of the action grid, confirming that the symmetric pure NE does not exist in this canonical setting.

foundational baseline on which such incentive schemes must rest: it characterises the equilibrium that emerges when no incentive is offered, which is what any incentive design needs to improve upon.

f) Utility-noise tradeoff in DP-SGD.: We motivate the functional form of the model-utility value $V(\sigma)$ via the excess-empirical-risk bound of [**?**] for convex, Lipschitz DP-ERM, who prove that the excess risk of DP-SGD scales as $O(d/(n\varepsilon)^2)$ at fixed δ, equivalently as $O(\sigma^2/n^2)$ under the moments accountant of [9]. The convex setting is restrictive; for deep learning, [**?**] provide extensive empirical evidence that test accuracy degrades monotonically with σ on standard benchmarks. We use $V(\sigma) = 1/(1+\sigma^2)$ as a directional proxy; the qualitative results of our paper do not depend on the precise functional form.

VII. Discussion and Conclusion

We have shown that sampling-based individual differential privacy is not just *vulnerable* to collusion attacks but *equilibrium-asymmetric*. The pure Nash equilibria of the Privacy Externality Game, when they exist on the dense action grid, sit *off the diagonal*: one player accepts a higher realised MIA advantage and is compensated by a higher model-utility benefit; the other absorbs the opposite trade. The budget-manipulation attack of [5] is precisely the asymmetric equilibrium played by a coalition that weights the target's harm in its joint utility, and Theorem 9 shows this is the rational best response of *any* coordinated subset of data holders.

a) Limitations of this work.: Three caveats are worth noting. First, our model assumes one-shot play. Real deployments of iDP are dynamic: contributors join and leave; budgets are re-negotiated; trainers re-train. A repeated-game extension is the natural next step, since the asymmetric equilibrium we observe could in principle be averaged out across rounds (or exacerbated, as a player who is repeatedly assigned the high-A role would exit the system). Second, we assume full information about other players' actions. Real contracts may keep budgets private, and Theorem 6 already distinguishes between fully aware players and players who only see their nominal (ε_i, δ) contract. A Bayesian PEG that treats ε_i as a private type drawn from a known distribution is a natural and important extension. Third, our analysis is restricted to the sampling-based realisation of iDP. The sensitivity-based realisation [2] adjusts per-sample clipping bounds rather than per-sample sampling rates and does not exhibit the same externality (the noise-to-sensitivity ratio is preserved by construction). Our framework specialises to that case as the trivial PEG in which the externality vanishes.

b) The role of the model-utility function V.: Our choice $V(\sigma) = 1/(1+\sigma^2)$ is motivated by the convex DP-ERM excess-risk bound of [**?**], and we use it as a directional proxy in our possibly non-convex setting. The qualitative findings—existence of asymmetric NEs, monotone coalition attack, externality lemma—do not depend on the precise shape of V; they require only that V is continuous, monotone-decreasing in σ, and bounded. We expect the same qualitative picture under any utility function with these three properties.

c) Mitigation.: The natural mitigation is to make the contract internalise the externality. The $(\varepsilon_i, \delta, \Delta)$-iDP formulation of [8] adds a Δ-divergence bound on the realised vulnerability, restricting the action space of the PEG to profiles that satisfy a per-record bound on excess advantage. We expect this restriction to admit a unique equilibrium and to drive the realised advantage gap between players to within Δ by definition. A clean characterisation of the constrained PEG is figures/fig_C2_asymmetric.pdf

Fig. 2: C2: equilibrium budget (left) and realised MIA advantage (right) as a function of dataset size ratio $|G_1|/|G_2|$, on the log-spaced action grid. Where multiple pure NEs exist, we report the one with the largest asymmetry in realised advantage.

left to future work.

d) Open questions.: The following questions naturally arise from this work but lie outside its scope:

1) Does a pure-strategy NE always survive on the continuous action space under the coupled cost, and does a Δ-bound on the slack suffice for existence?
2) Is the PEG a potential game in the sense of [15]? PoP's analysis was a potential game because the privacy cost was separable; the externality term breaks this directly. Conditions on the cross-partial of A_i that recover the potential structure would unify the two settings.
3) What is the sign of the externality across composition regimes (minority vs majority weak-privacy)? Our experiments suggest the smaller group is favoured, but the analytic dependence on $|G_g|$ deserves a closed-form characterisation.

figures/fig_C4_coalition.pdf

Fig. 3: C4: realised MIA advantage of the target as a function of coalition size k, with the coalition playing the participation-preserving minimum budget at each k. The target's vulnerability rises monotonically and exceeds $A_t = 0.7$ for $k = 4$.

4) What is the appropriate player granularity (single sample versus silo) in the FL setting, and how does the granularity change the equilibria?
5) Worst-case (analytical) vs realised (empirical) cost—which should the contract be written against, and how large is the gap in practice across deep-learning workloads?
6) Is the $(\varepsilon_i, \delta, \Delta)$-iDP contract by itself sufficient as a collusion-resistant mechanism, or are sensitivity-based iDP or public-data padding needed when the slack is exceeded?

e) Wider lesson.: The wider lesson is that an *individual* privacy contract is incomplete: it does not pin down the collective dynamics. Where individuals interact strategically (in a single machine-learning training, in a federated round, in any future composition of the iDP mechanism), the contract must also constrain their joint behaviour or the equilibrium will quietly violate at least one participant's expectations while preserving every formal guarantee.

f) Conclusion.: The promise of iDP—that every data contributor controls her own privacy—is broken by the privacy externality of sampling-based mechanisms. We formalised the strategic situation faced by iDP data contributors as the Privacy Externality Game, proved monotonicity of the externality and characterised the equilibria. The result is that the collusion attack on iDP is not adversarial but *rational*, and the remedy must operate at the level of the contract.

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figures/fig_C5_surface.pdf

Fig. 4: C5: target's realised MIA advantage as a function of the other group's ε and dataset proportion, for two target budgets $\varepsilon_T \in \{4, 16\}$. The curves rise as ε_O falls and as the proportion of the other group grows.

Appendix A
Detailed mechanism-solver construction

We elaborate on the bisection used to obtain the mechanism solution $(\sigma, p_1, \ldots, p_n)$ for an action profile $\boldsymbol{\varepsilon}$.

a) Inner step: p_g from (σ, ε_g).: For each fixed σ and each group g, we want the largest $p_g \in [0, 1]$ satisfying the I-fold (p_g, σ)-subsampled Gaussian at (ε_g, δ)-DP under (C1). Since the ε value of the subsampled Gaussian is strictly increasing in p_g (more sampling $\Rightarrow$ more privacy loss), a unique $p_g \in (0, 1]$ exists, capped at 1 if the budget cannot be exhausted. We compute the per-group ε by Rényi-DP composition [16] and binary-search p_g to bring this ε to ε_g to within a relative tolerance of 10^{-3}.

b) Outer step: σ from $(p_g, \boldsymbol{\varepsilon})$.: Define $B(\sigma) = \sum_g |G_g|\, p_g(\sigma, \boldsymbol{\varepsilon})$ as the realised expected batch size. Since each p_g is monotone-increasing in σ, so is B. We bisect σ to bring B to the target E_b, again to within a relative tolerance of 10^{-3}.

c) Participation threshold.: We declare profile infeasible (per Theorem 4) if any p_g falls below 10^{-5}. Below this threshold the group's contribution to (C2) is numerically negligible and the iDP contract is trivially satisfied. We verified the qualitative results are insensitive to the choice of this threshold across $[10^{-6}, 10^{-3}]$.

d) MIA advantage.: Given the mechanism solution (σ, p_g), we compute the MIA advantage of group g exactly from the privacy-loss-distribution of the I-fold (p_g, σ)-subsampled Gaussian using [11], then evaluate the trade-off function and Eq. (2). This is the same formulation used by [5] and yields the tightest analytical MIA advantage for the resulting mechanism.

APPENDIX B
VALIDITY OF $V(\sigma) = 1/(1 + \sigma^2)$

a) Frame.: The convex-DP-ERM excess-risk bound of [**?**] is, for a 1-Lipschitz L-strongly-convex $\ell\colon \mathcal{X} \times \mathcal{D} \to \mathbb{R}$,

$$\mathbb{E}[\hat{L}(\widehat{\theta}) - \hat{L}(\theta^*)] = O\Big(\frac{d \log(1/\delta)}{n^2 \varepsilon^2}\Big).$$

Translating to noise multiplier via the moments accountant of [9], this reads $O(\sigma^2 \cdot d/n^2)$ in *excess* empirical risk. Treating "model utility" as −"excess risk" and re-normalising to $V \in [0, 1]$ produces $V \propto 1/(1 + \alpha\sigma^2)$ for a problem-dependent α. We set $\alpha = 1$ for simplicity; sensitivity analysis (Section C) shows the qualitative results are unchanged for $\alpha \in [0.1, 10]$.

b) Beyond convex.: For non-convex deep-learning losses no analogous closed-form bound is known. The empirical literature [**?**], [2], [9] reports monotone test-accuracy degradation with σ in standard benchmarks. We use $V(\sigma) = 1/(1+\sigma^2)$ as a *directional* proxy; the qualitative results of our paper (existence of asymmetric NEs, monotone coalition attack, externality lemma) do not depend on the precise shape of V; they require only that V is continuous, monotone-decreasing in σ, and bounded.

APPENDIX C
SENSITIVITY TO GRID RESOLUTION AND TO V

a) Grid refinement.: For the 2-player setting of Fig. 1 we re-ran the sweep with $K = 44$ log-spaced points (twice the resolution) and verified that the set of pure NEs is contained, modulo the new intermediate grid points, in the set found at $K = 22$. Specifically, the asymmetric NEs at $K = 22$ persist at $K = 44$ and the new intermediate grid points do not introduce additional NEs.

b) Functional form of V.: The same 2-player sweep was repeated with $V(\sigma) = e^{-\sigma}$ (an exponential decay alternative); the pure NEs shift quantitatively (mostly toward higher ε values, since $V_{\exp}$ falls off faster than $V_{1/(1+\sigma^2)}$ as σ grows) but the asymmetric structure is preserved.

c) Participation threshold.: Lowering the threshold to 10^{-6} extends the feasible region; the NEs found at 10^{-5} remain NEs but additional ones appear in the newly feasible interior. Raising the threshold to 10^{-3} shrinks the feasible region and removes some NEs at the boundary; the surviving NEs do not move.